

University of Hyderabad
School of Computer & Information Sciences

Database Management Systems

MCA – CA514 (3 credits) (3-0-0)

DBMS Lab

MCA – CA516 (1.5 credits) (0-0-3)

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Outline

- Prerequisites
- Course outcomes
- Course outline
- Grading

Prerequisites

- Should have good understanding of basic computer science concepts

- Would like to know the background of the students!

Course Outcomes (COs)

- CO-1: Understand the basic concepts of database management systems (Understand)
- CO-2: Design the database systems using ER and EER Models (Create)
- CO-3: Use SQL to query the database systems (Apply)
- CO-4: Evaluate the database design aspects by considering normalization principles (Evaluate)
- CO-5: Explain indexing and hashing mechanisms used in database systems (Understand)
- CO-6: Understand the transaction management in database systems (Understand)

Course outline

- UNIT-1
  - Introduction: Data and Information, Overview of File Management System, Database Systems Versus File Systems, Data Models, Database Languages- Data Definition Language, Data Manipulation Language, Functions of Database Administrator, Taxonomy of Database Users.
- UNIT-2
  - Data Models: Entity Relationship model- Basic Concepts, Constraints, Keys, Design Issues, E-R Diagram, Extended E-R Features, Design of E-R Database Schema, Reduction of E-R Schema to tables.
  - Relational Model- Structure of Relational Database, Normalization, Functional Dependencies, Decomposition, Desirable properties of Decomposition, Normal Forms, Denormalization for Performance
- UNIT-3
  - Structured Query Language (SQL). Basic structure, Set operations, Aggregate functions, Null Values, Nested Subqueries, Views, Complex queries, joins Modification of database schema, Integrity Constraints. Relational Algebra & Relational Calculus.

Course outline - continued

- UNIT-4
  - Application Design & Development:
  - Discussion on complex data types
    - Semi structured data, object-based data, temporal data and spatial data
  - Big data analytics and how those applications place distinct demand on data management compared with traditional database applications
    - NoSQL, MapReduce, Apache Spark, streaming data and graph databases
- UNIT-5
  - Storage and File Structures:
    - File organization, Organization of Records in Files, Data Dictionary Storage.
  - Indexing and Hashing:
    - Basic Concepts, Ordered Indices, Static Hashing, Dynamic Hashing, Comparison of Ordering Indexing and Hashing, Index Definition in SQL, Multiple-Key Access
  - Transaction Management:
    - ACID properties, Transaction State, Concurrent Executions, Serializability, Recoverability, Transaction Definition in SQL, Concurrency Control – Lock-Based Protocols, Timestamp-Based Protocols

Schedule

- Lecture
  - Monday 14:00 – 16:00
  - Thursday 14:00 – 15:00
- Lab / practice session
  - Friday 10:00 – 13:00

Grading

- Most weeks there will be an exercise
  - To be submitted during the lab or within a week (based on the task)
  - Late submissions, if deadline is a week
    - 10% will be deducted per day
    - You can submit for 50% evaluation up to one week delay or under unavoidable circumstances (with prior agreement)
- For the lab part may give bigger exercises (Mini-projects) with clear deadlines
- Best of 2 minors for theory part
- Final examinations
  - Both for theory and lab

References & Books

- Moodle course links

DBMS: https://scislearn2.uohyd.ac.in/moodle/course/view.php?id=258

- A. Silberschatz, H. F. Korth and S. Sudarshan, Database System Concepts, 7th Edition McGraw Hill Publications
- R. Elmasri, S. B. Navathe: Fundamentals of Database Systems, 7th Edition, Pearson Publication.